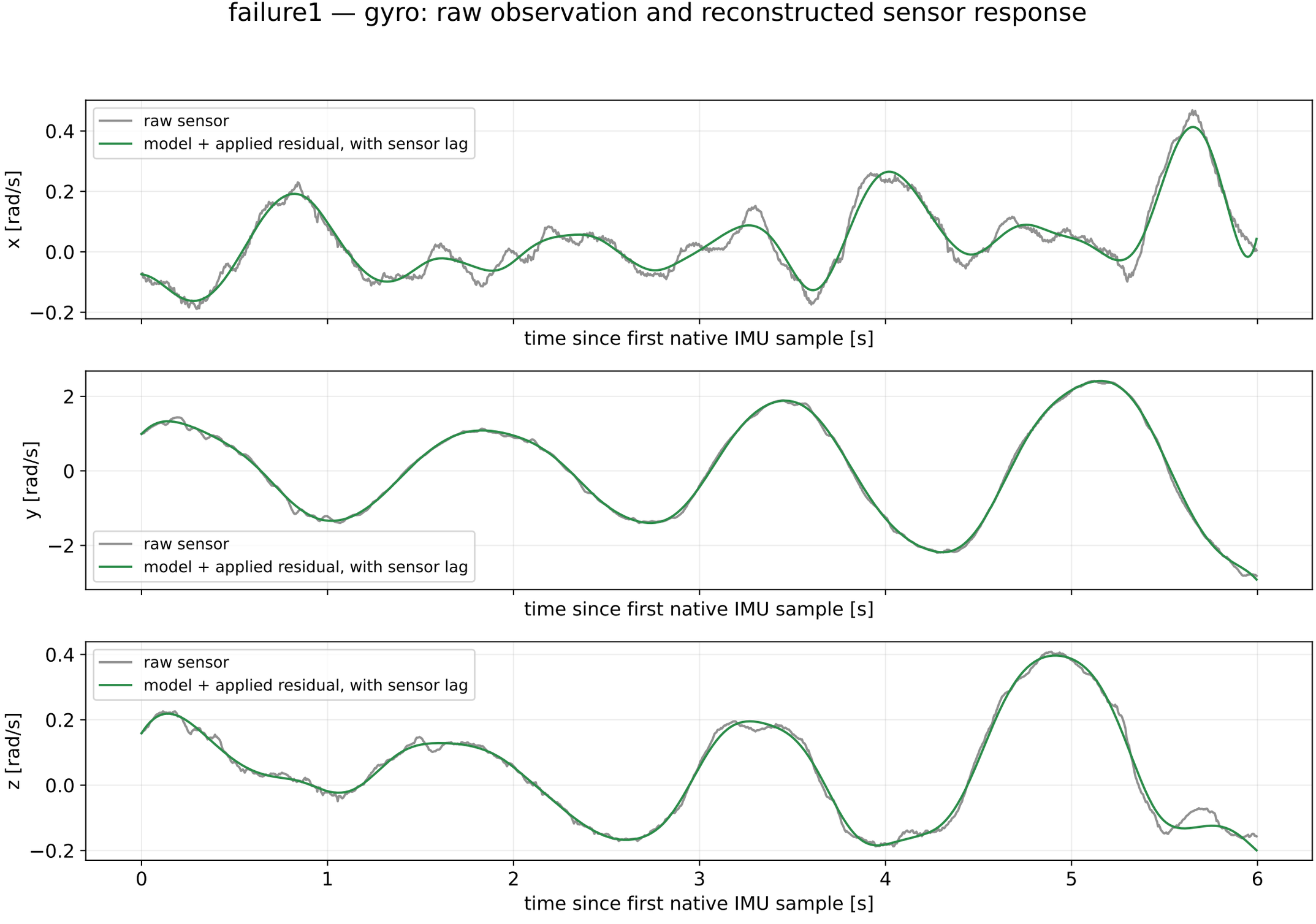
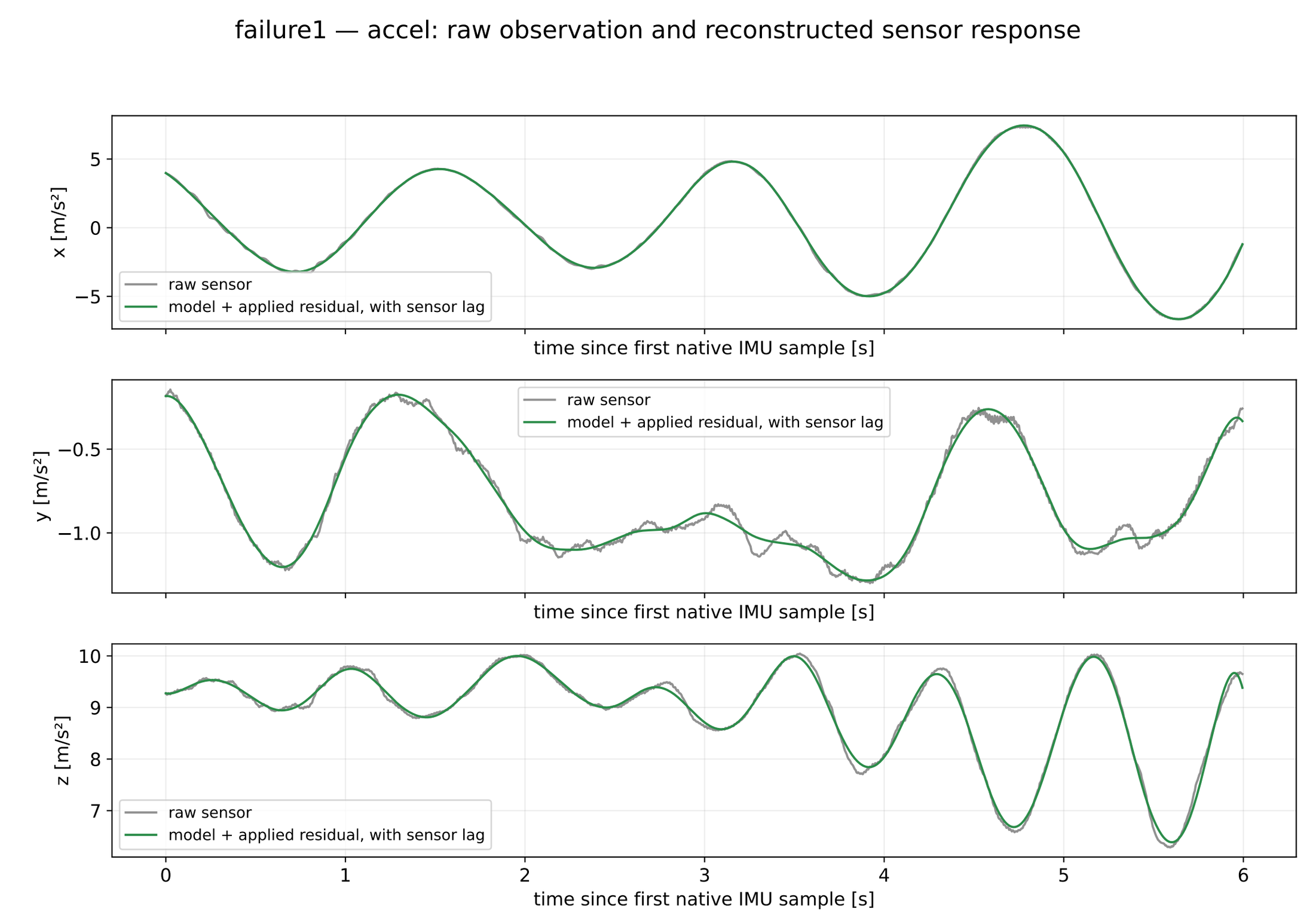
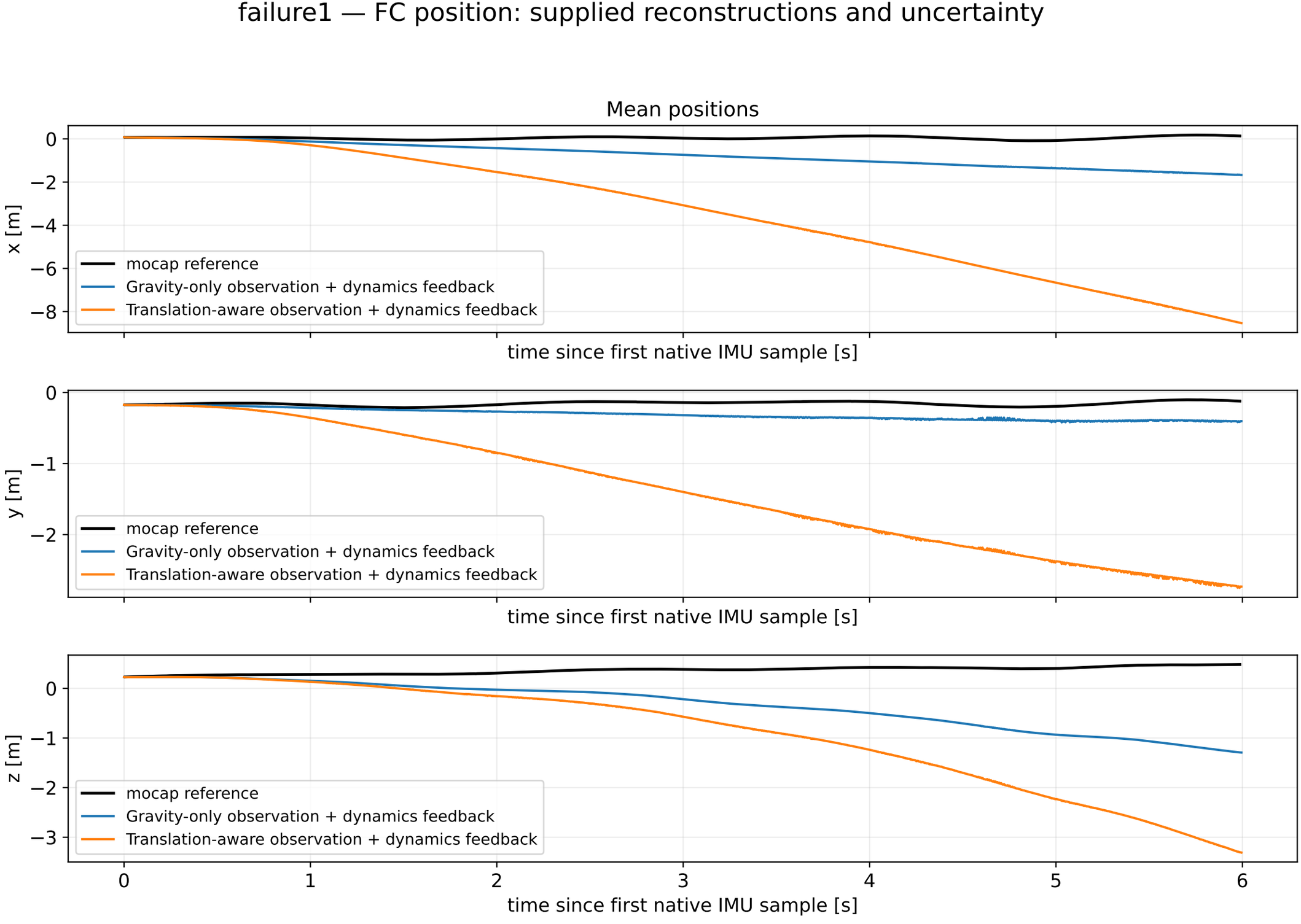


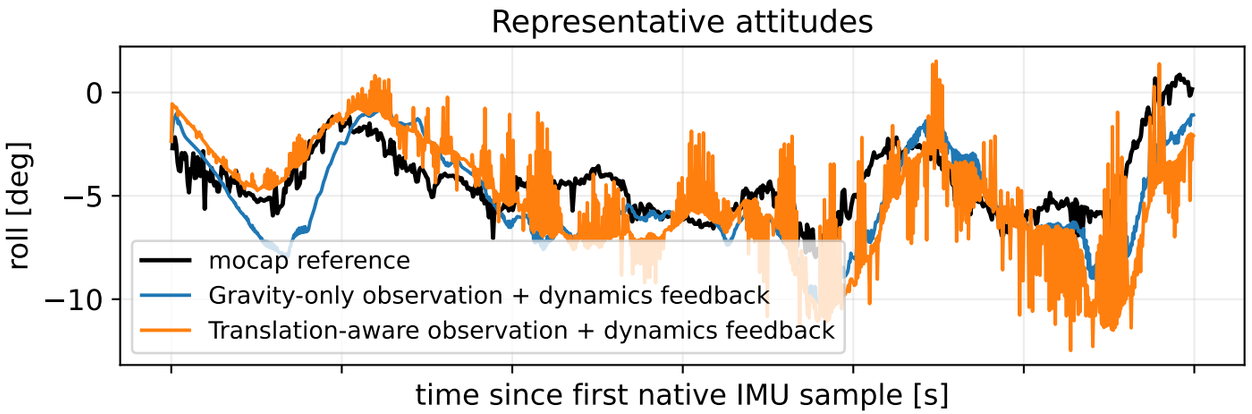
τ = 0.0371901 s; forward response τ ẏ + y = physical model input. Native-interval ZOH; initialized by the first raw reading.

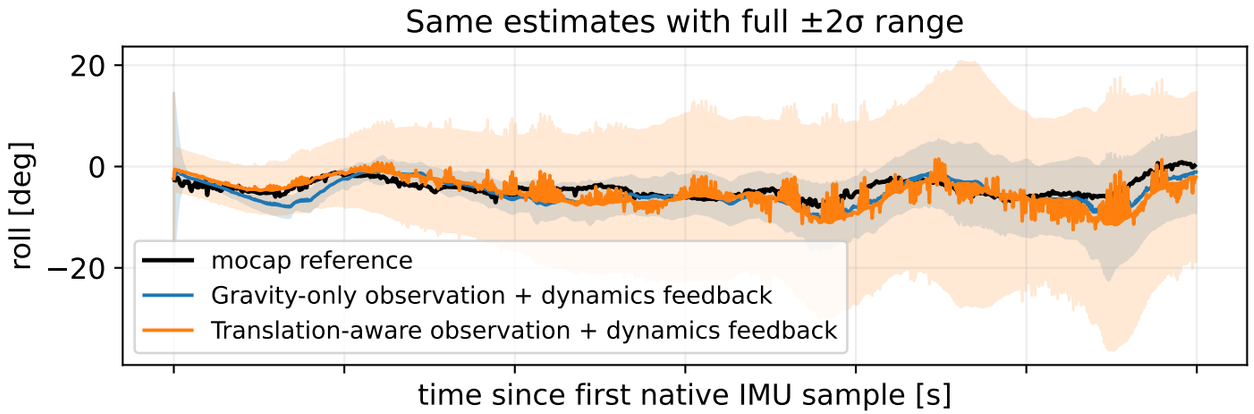

failure1 — accel: raw observation and reconstructed sensor response
x [m/s²]
y [m/s²]
z [m/s²]
raw sensor
model + applied residual, with sensor lag
time since first native IMU sample [s]
τ = 0.0662059 s; forward response τ ẏ + y = physical model input. Native-interval ZOH; initialized by the first raw reading.

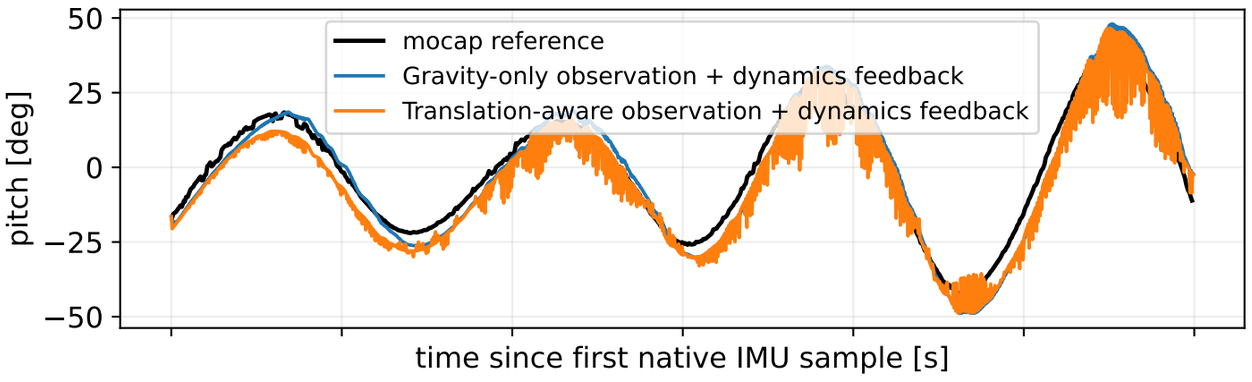


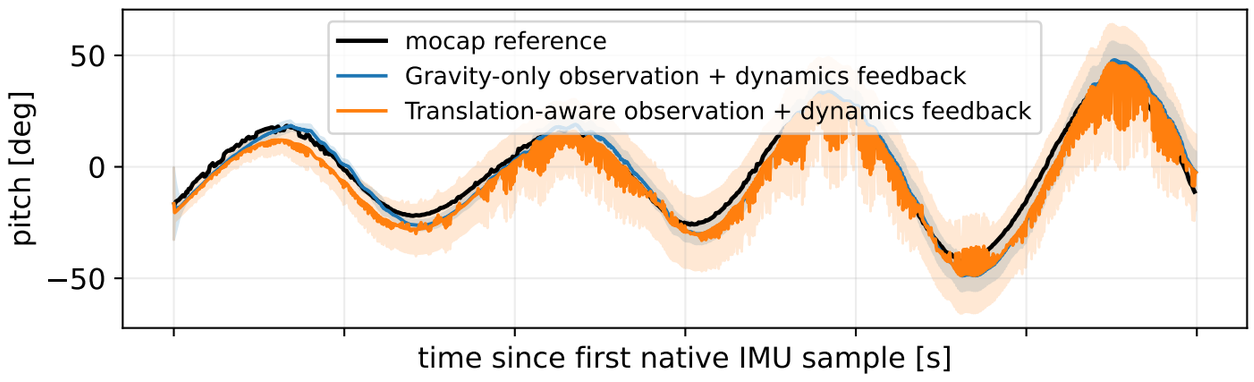
Model position covariance bands omitted by request; computed covariance data retained.
black reference uses original mocap stamps.

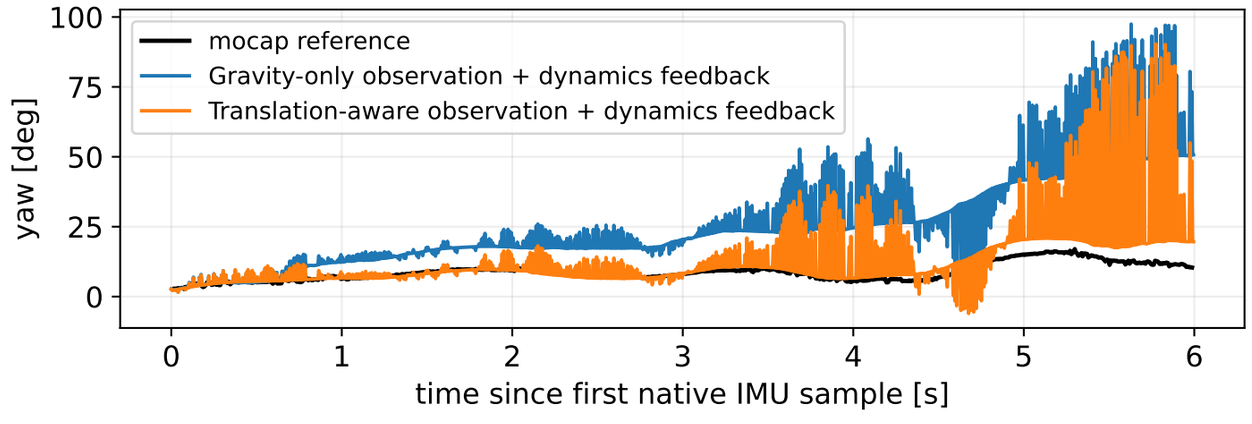
failure1 — attitude: supplied reconstructions and uncertainty

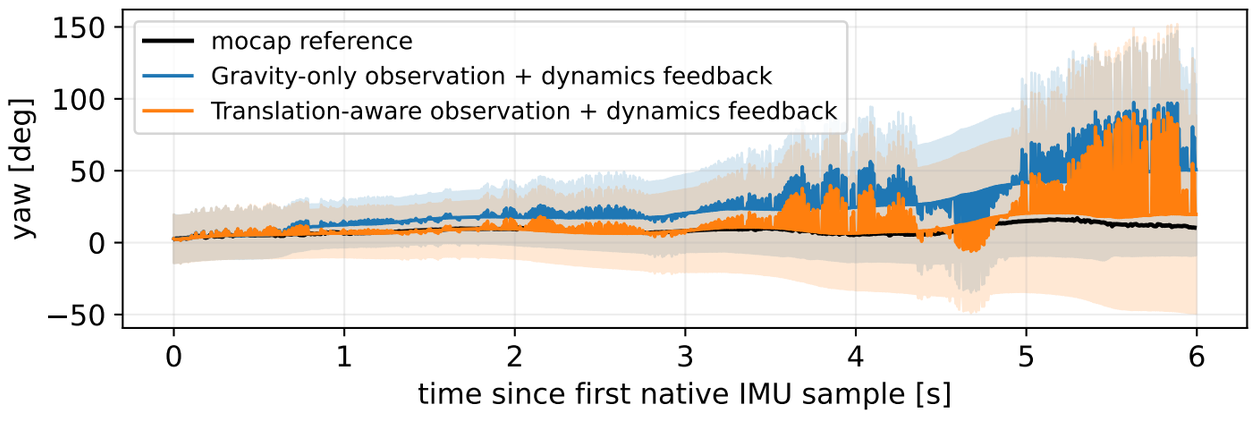




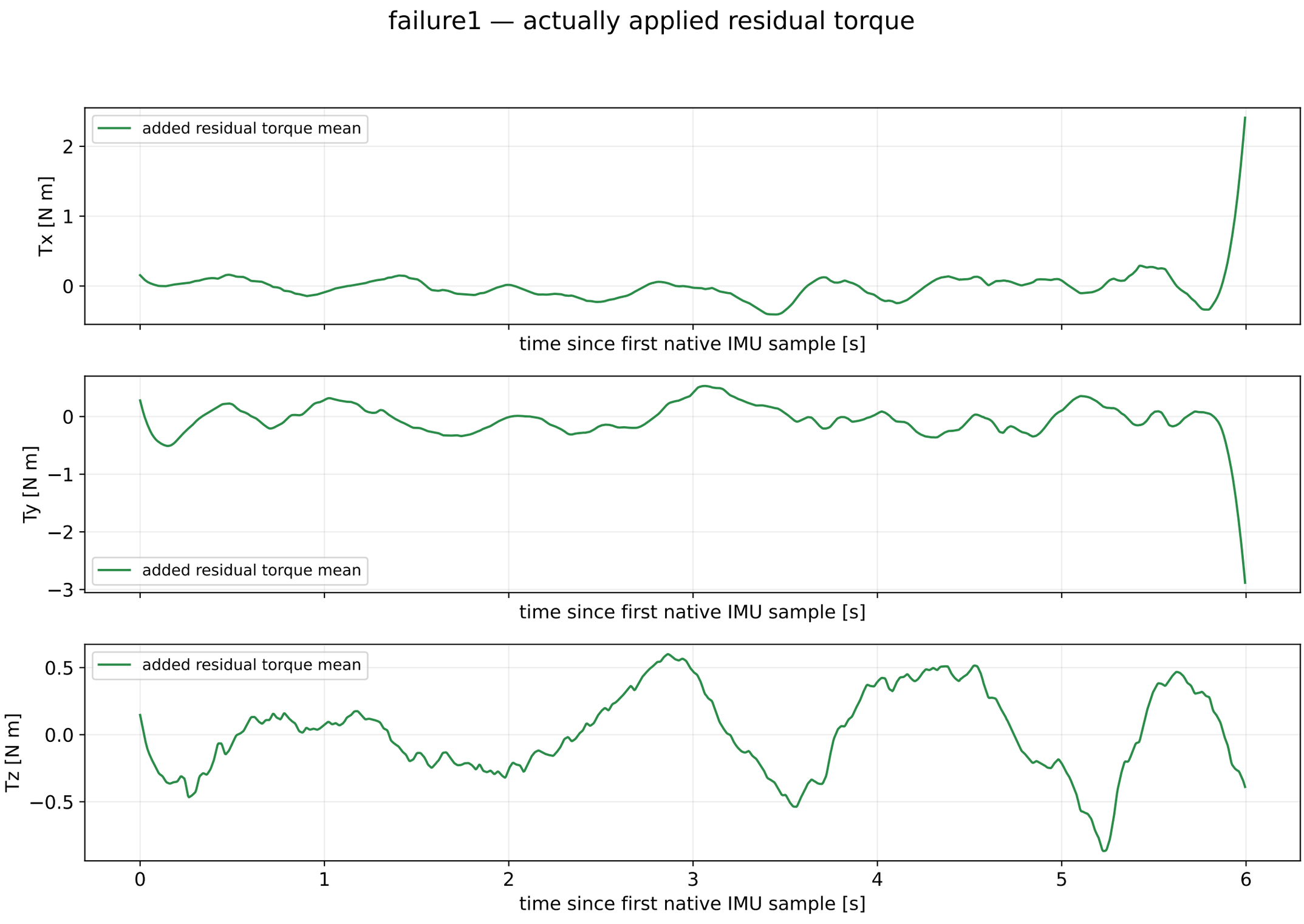




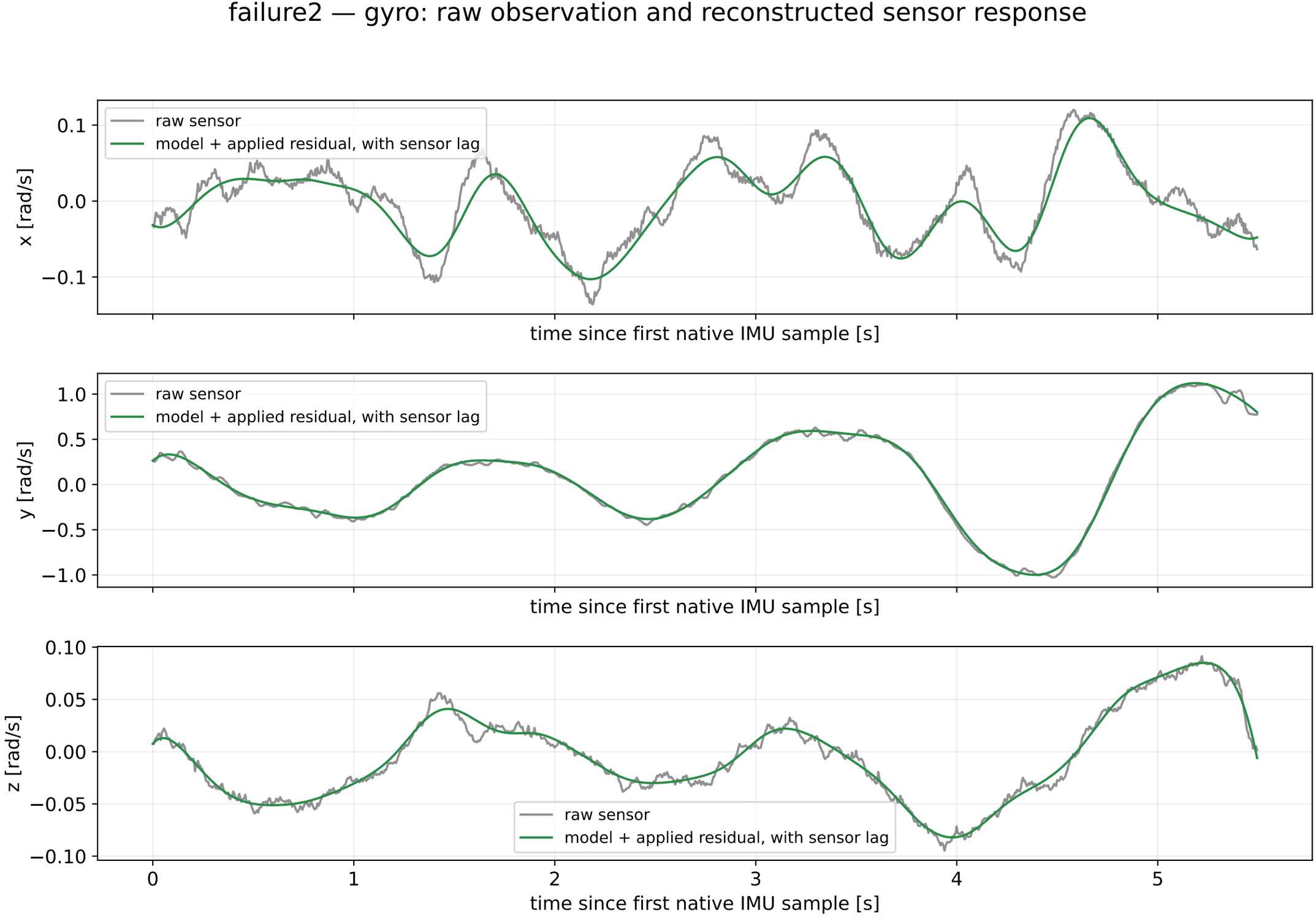




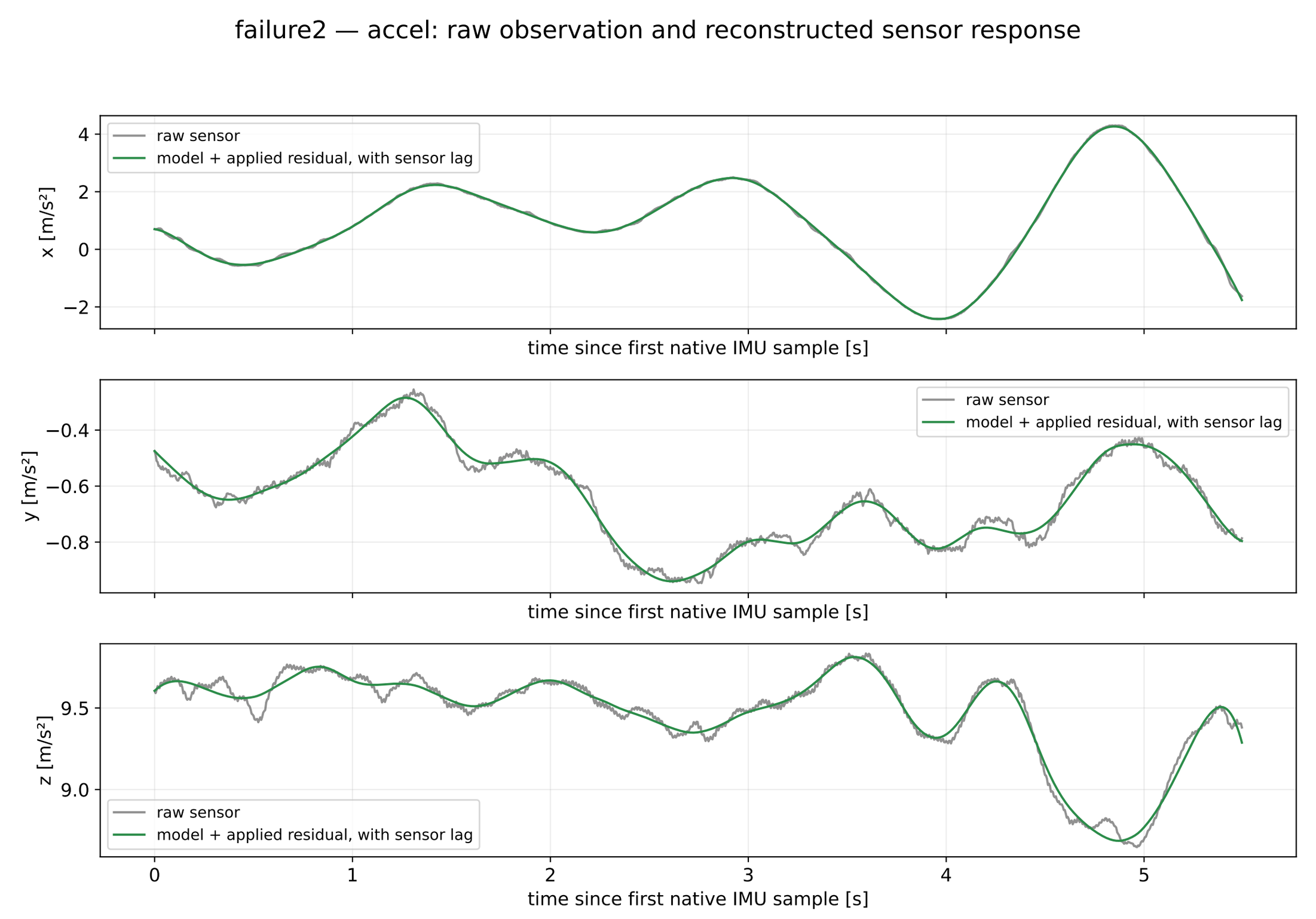
Attitude: IMU posterior-law local Euler approximation.
±2σ is not a 95% guarantee. Right column retains complete bands; black reference uses original mocap stamps.



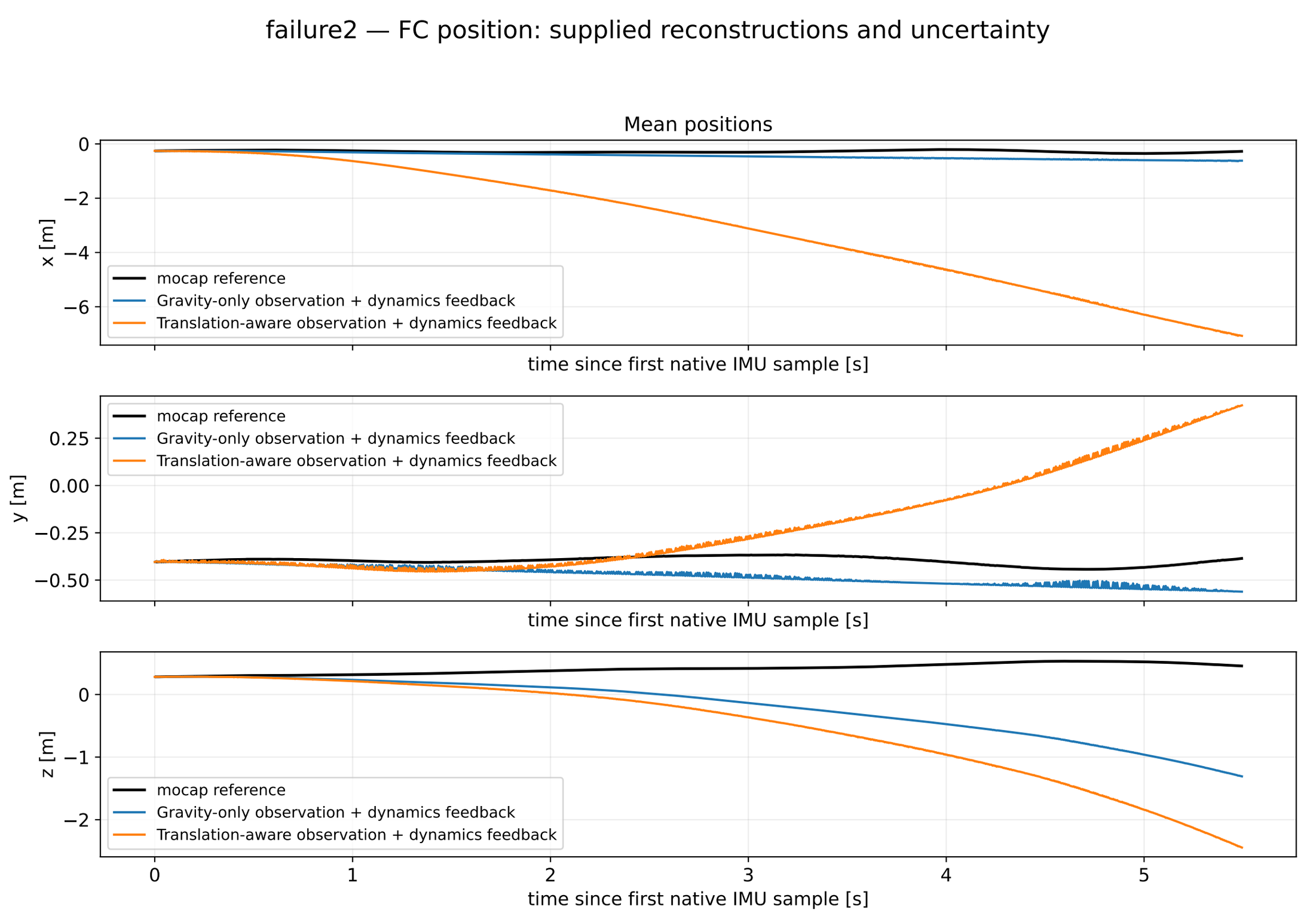
Body coordinates, about the center of gravity; the same torque mean added to reconstruction dynamics.



τ = 0.0615987 s; forward response τ ẏ + y = physical model input. Native-interval ZOH; initialized by the first raw reading.

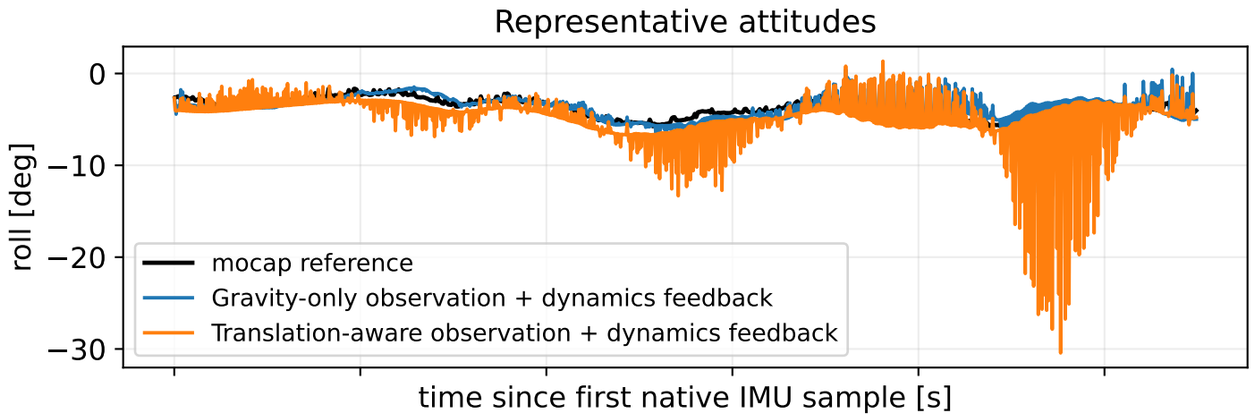


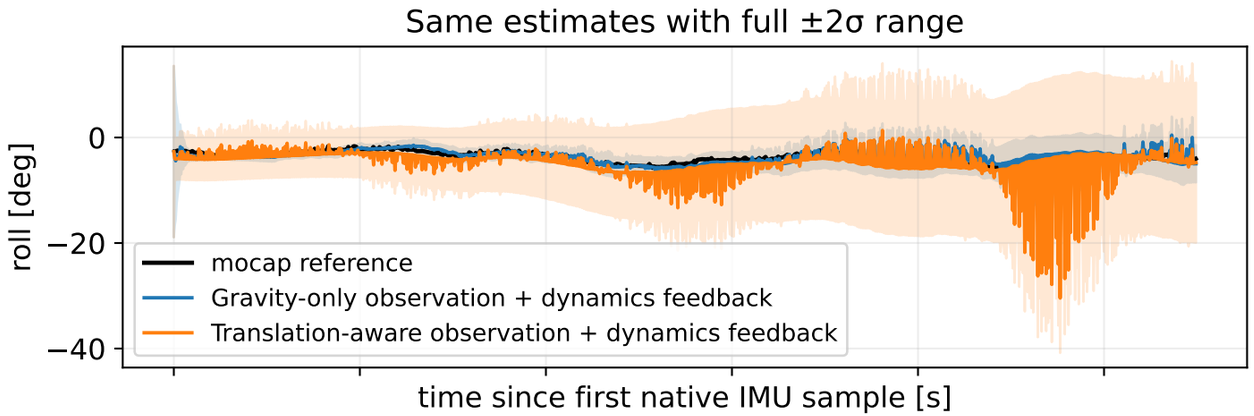
τ = 0.0620911 s; forward response τ ẏ + y = physical model input. Native-interval ZOH; initialized by the first raw reading.

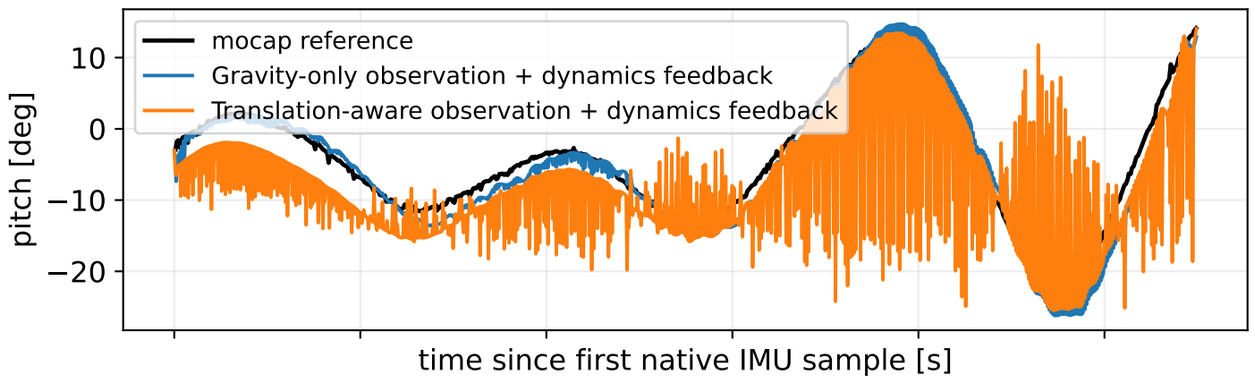


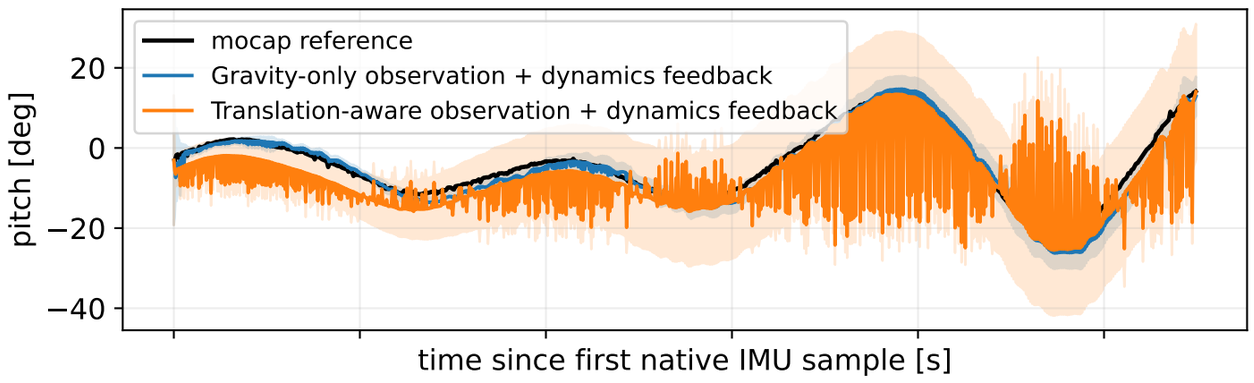
Model position covariance bands omitted by request; computed covariance data retained.
black reference uses original mocap stamps.

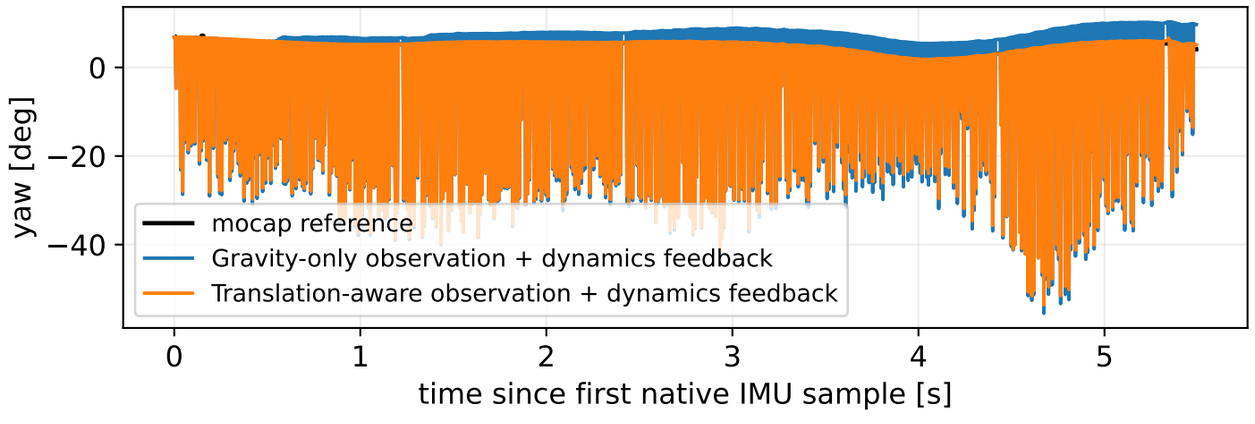
failure2 — attitude: supplied reconstructions and uncertainty

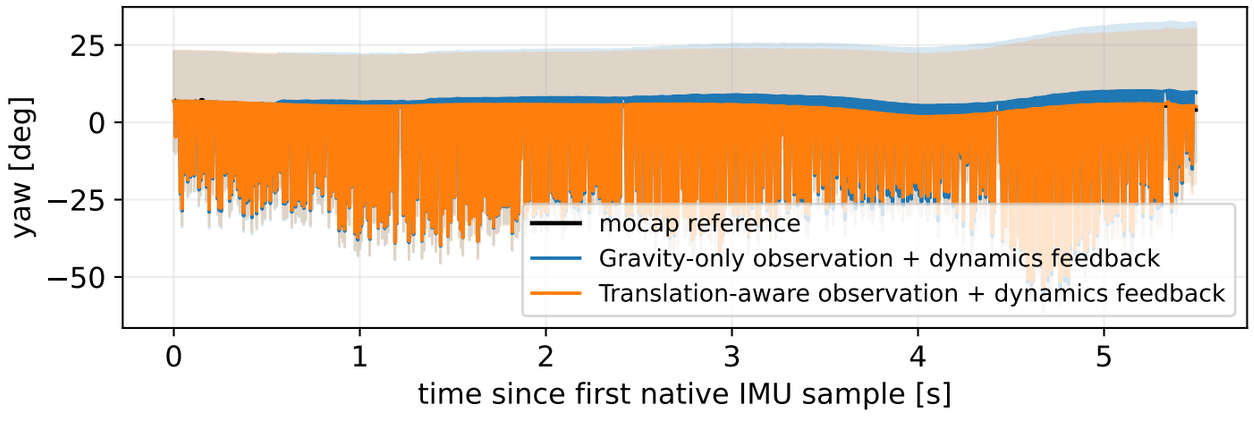




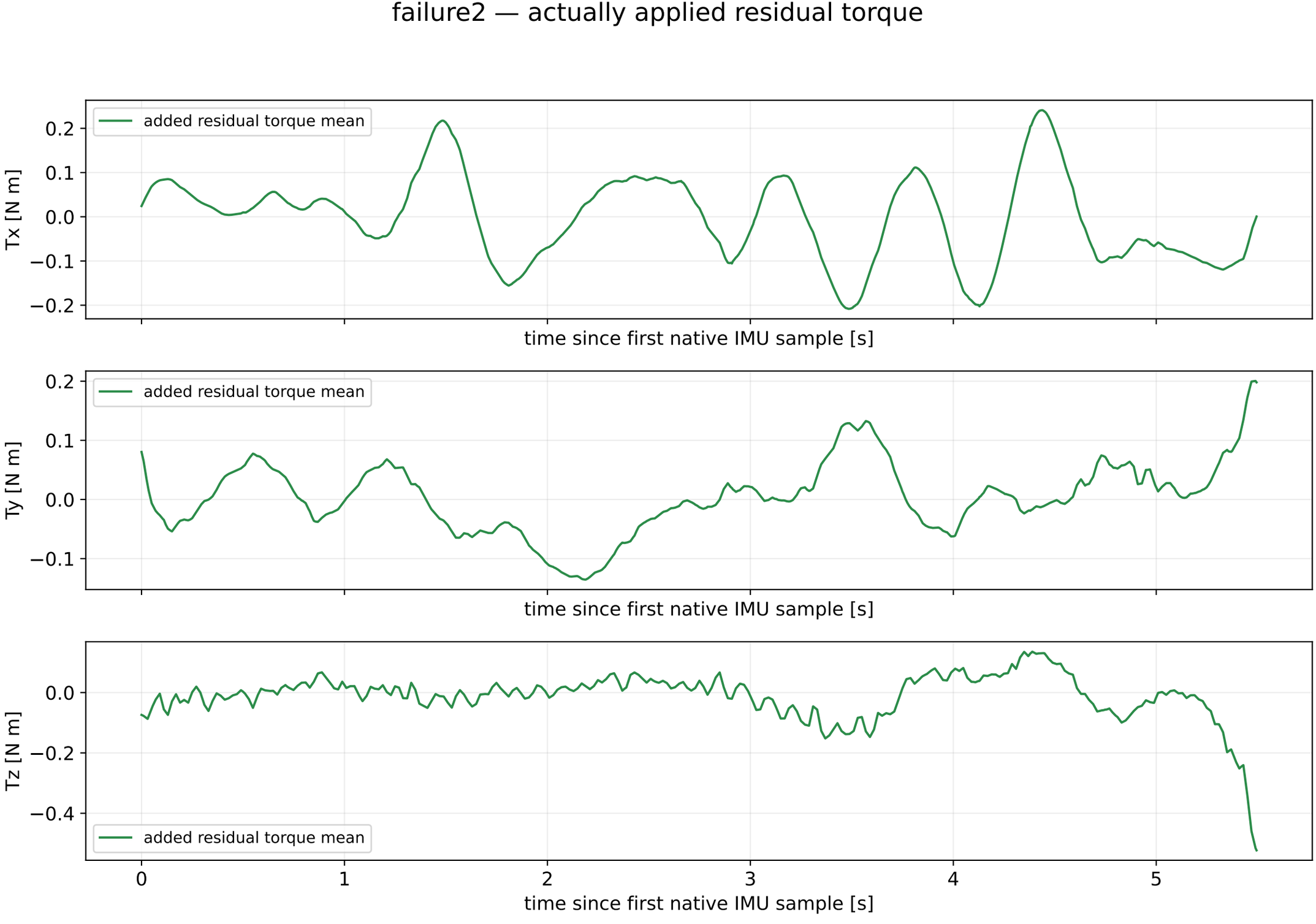




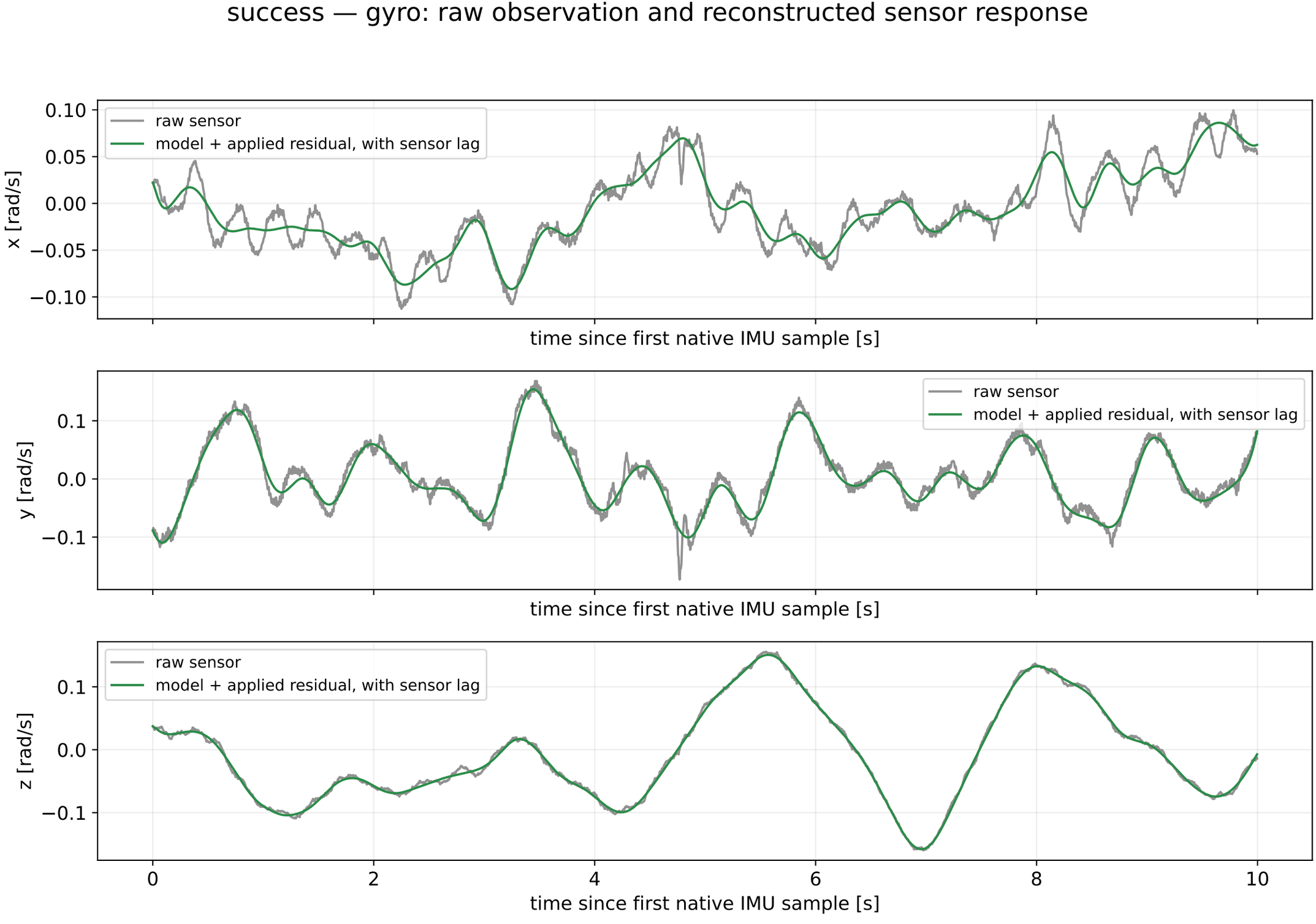


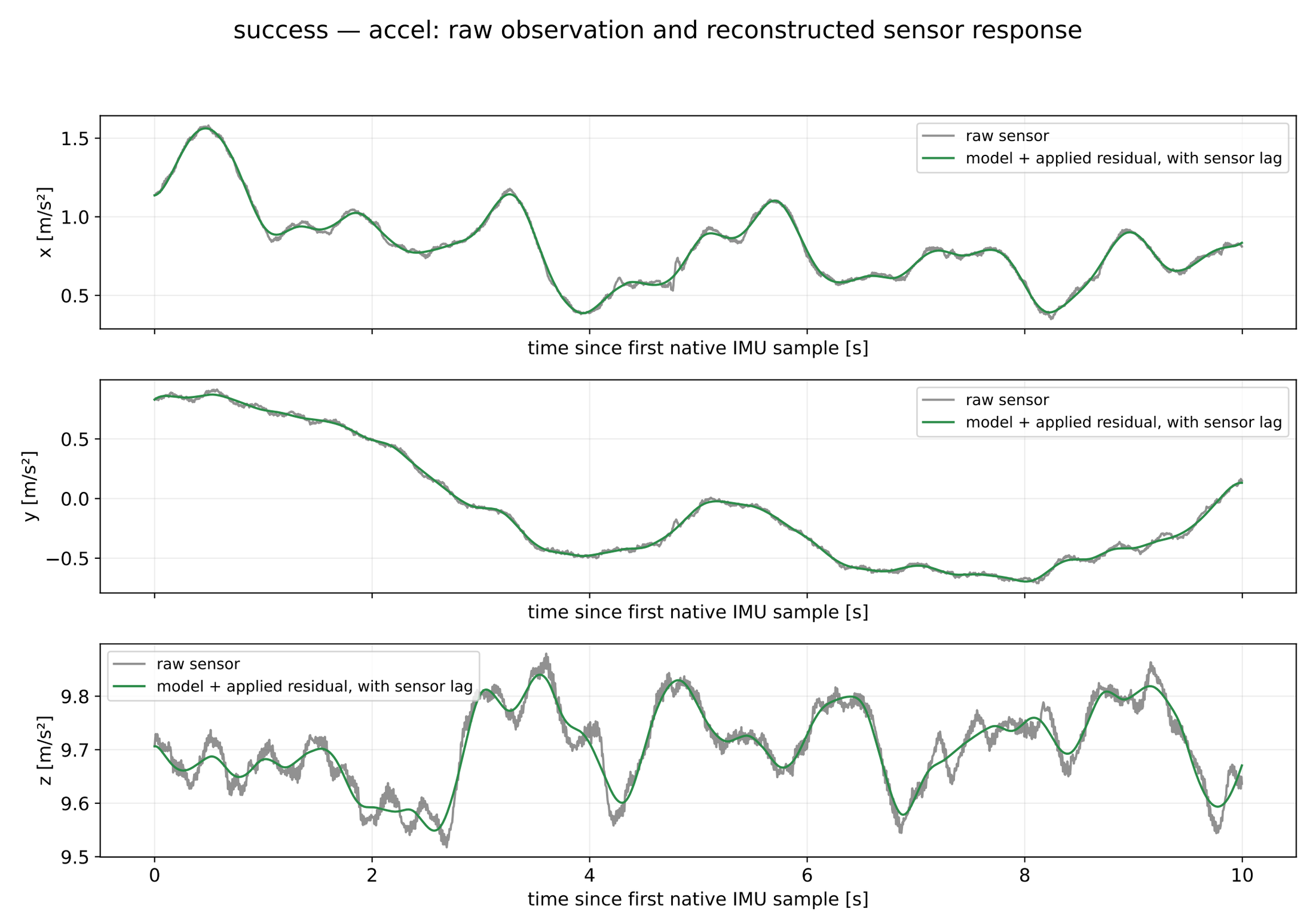


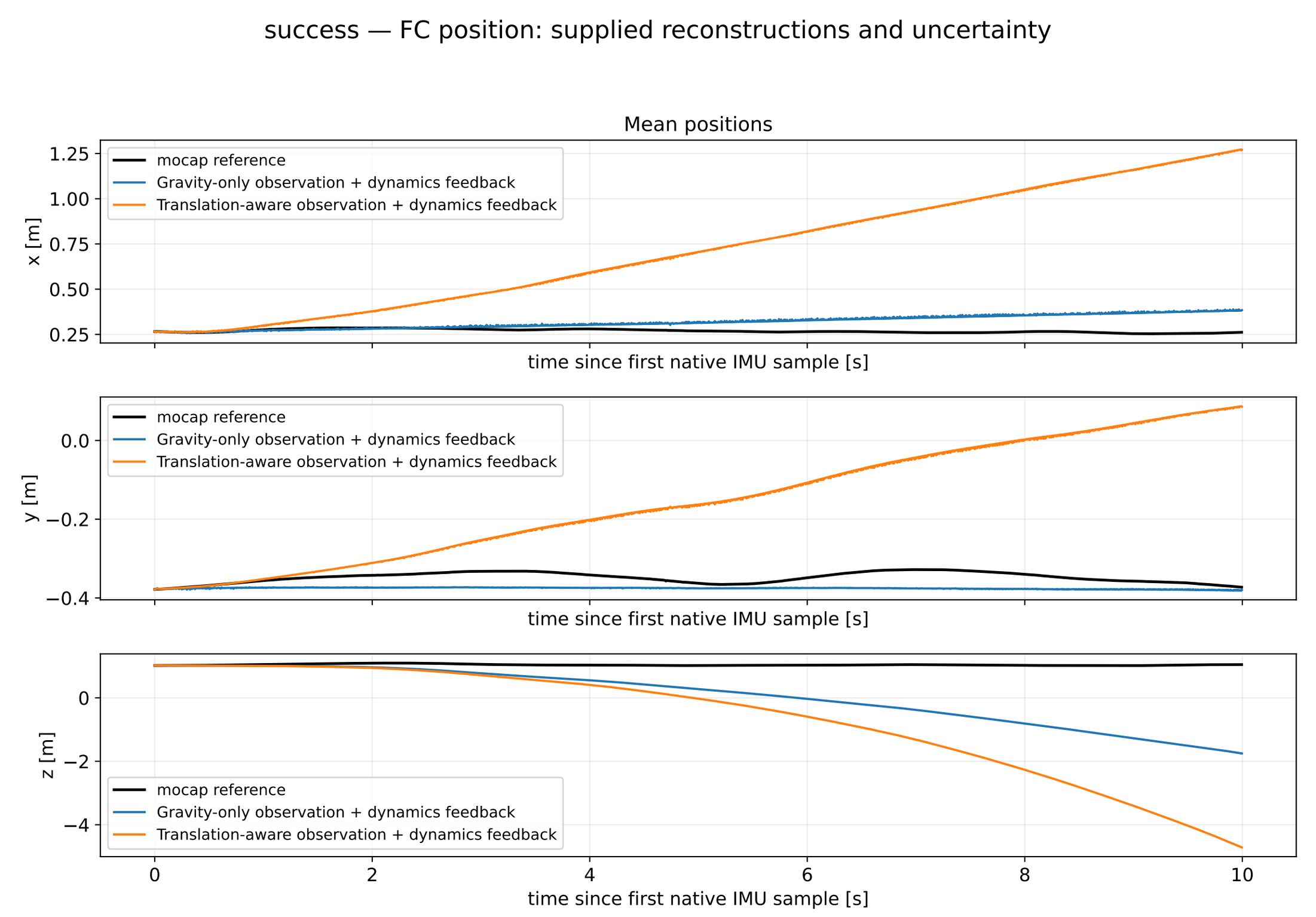
Attitude: IMU posterior-law local Euler approximation.
±2σ is not a 95% guarantee. Right column retains complete bands; black reference uses original mocap stamps.



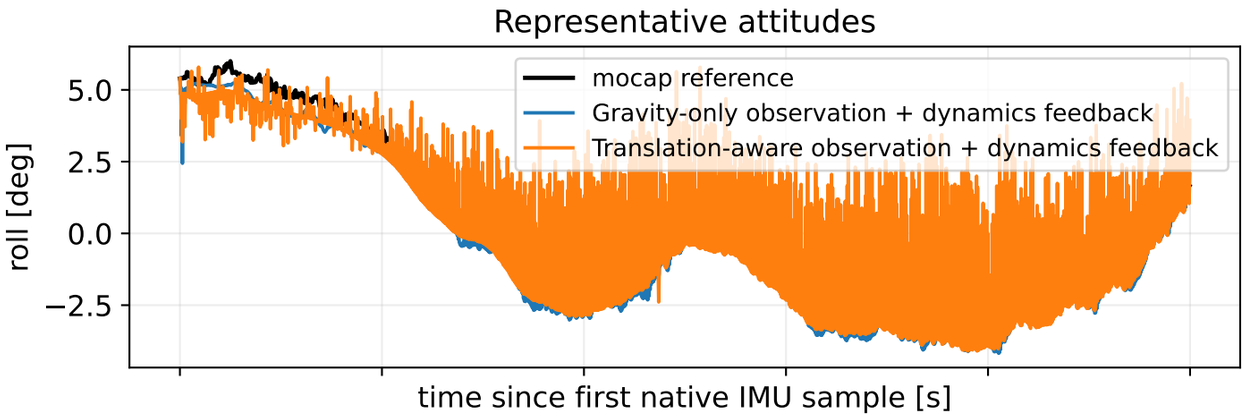
Body coordinates, about the center of gravity; the same torque mean added to reconstruction dynamics.

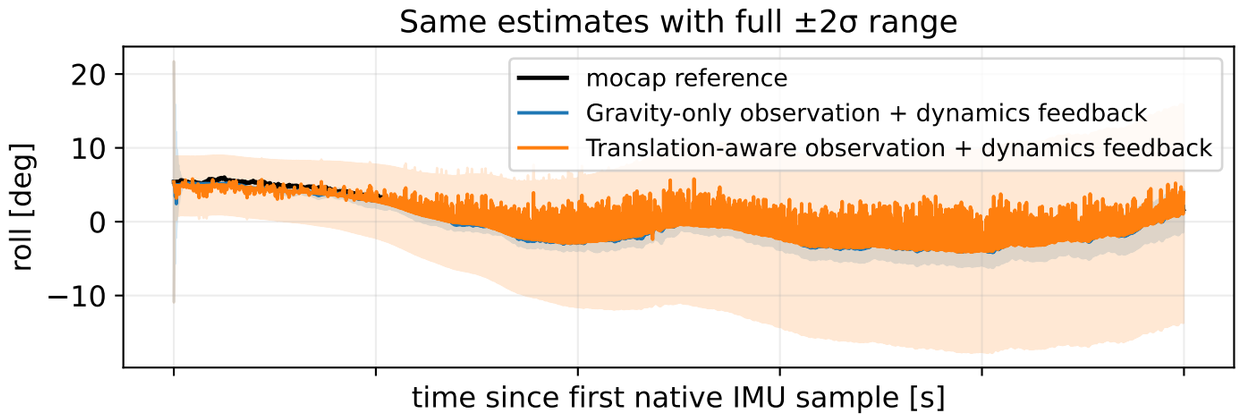

success — gyro: raw observation and reconstructed sensor response
raw sensor
model + applied residual, with sensor lag
x [rad/s]
0.10
0.05
0.00
−0.05
−0.10
time since first native IMU sample [s]
raw sensor
model + applied residual, with sensor lag
y [rad/s]
0.1
0.0
−0.1
time since first native IMU sample [s]
raw sensor
model + applied residual, with sensor lag
z [rad/s]
0.1
0.0
−0.1
0
2
4
6
8
10
time since first native IMU sample [s]
τ = 0.0309382 s; forward response τ ẏ + y = physical model input. Native-interval ZOH; initialized by the first raw reading.

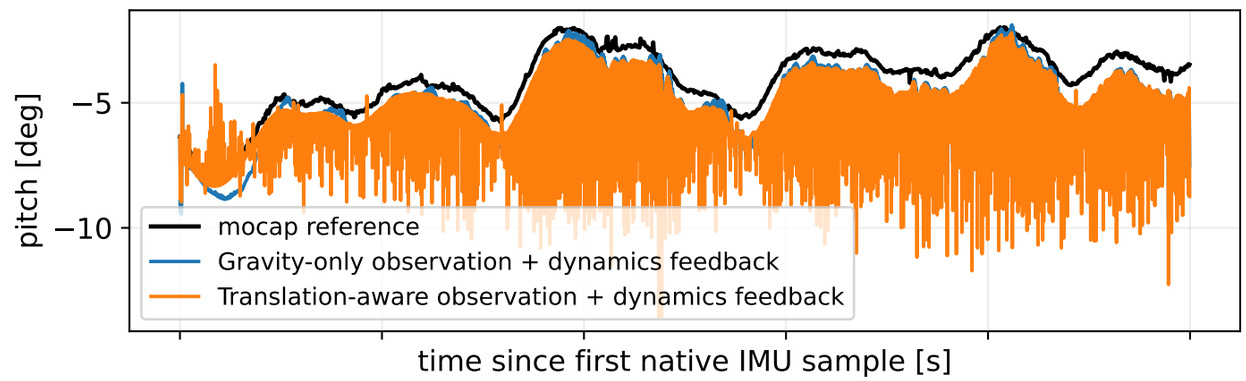

success — accel: raw observation and reconstructed sensor response
raw sensor
model + applied residual, with sensor lag
x [m/s²]
1.5
1.0
0.5
time since first native IMU sample [s]
raw sensor
model + applied residual, with sensor lag
y [m/s²]
0.5
0.0
−0.5
time since first native IMU sample [s]
raw sensor
model + applied residual, with sensor lag
z [m/s²]
9.8
9.7
9.6
9.5
0
2
4
6
8
10
time since first native IMU sample [s]
τ = 0.06319 s; forward response τ ẏ + y = physical model input. Native-interval ZOH; initialized by the first raw reading.

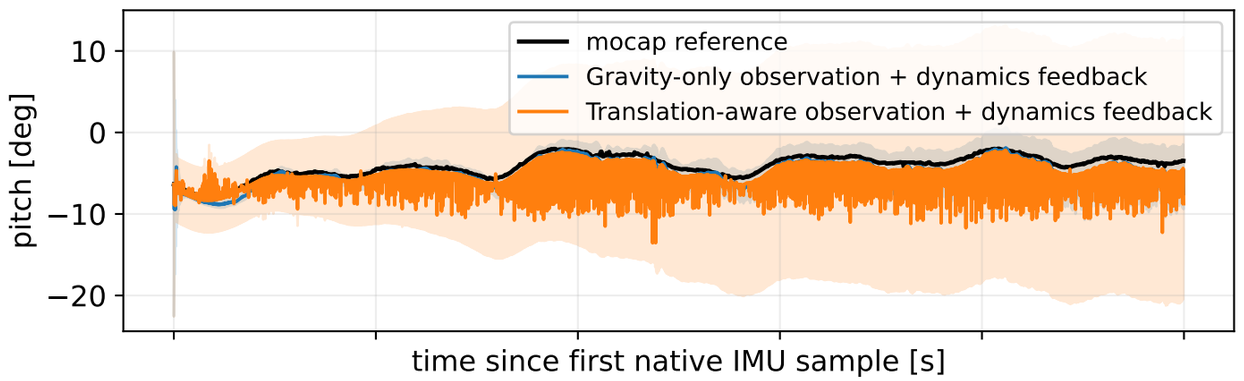

success — FC position: supplied reconstructions and uncertainty
Mean positions
mocap reference
Gravity-only observation + dynamics feedback
Translation-aware observation + dynamics feedback
x [m]
1.25
1.00
0.75
0.50
0.25
time since first native IMU sample [s]
mocap reference
Gravity-only observation + dynamics feedback
Translation-aware observation + dynamics feedback
y [m]
0.0
−0.2
−0.4
time since first native IMU sample [s]
mocap reference
Gravity-only observation + dynamics feedback
Translation-aware observation + dynamics feedback
z [m]
0
−2
−4
0
2
4
6
8
10
time since first native IMU sample [s]
Model position covariance bands omitted by request; computed covariance data retained.
black reference uses original mocap stamps.

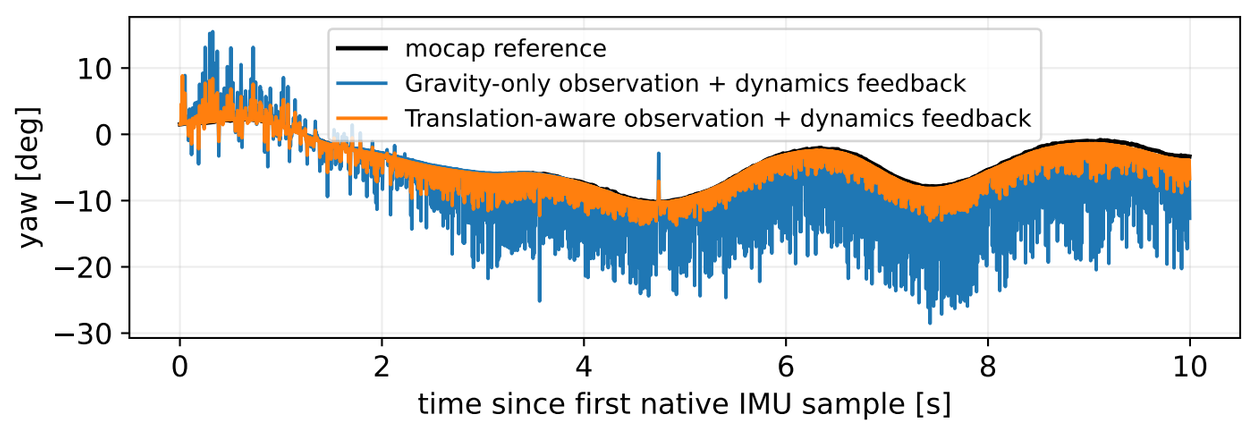
success — attitude: supplied reconstructions and uncertainty

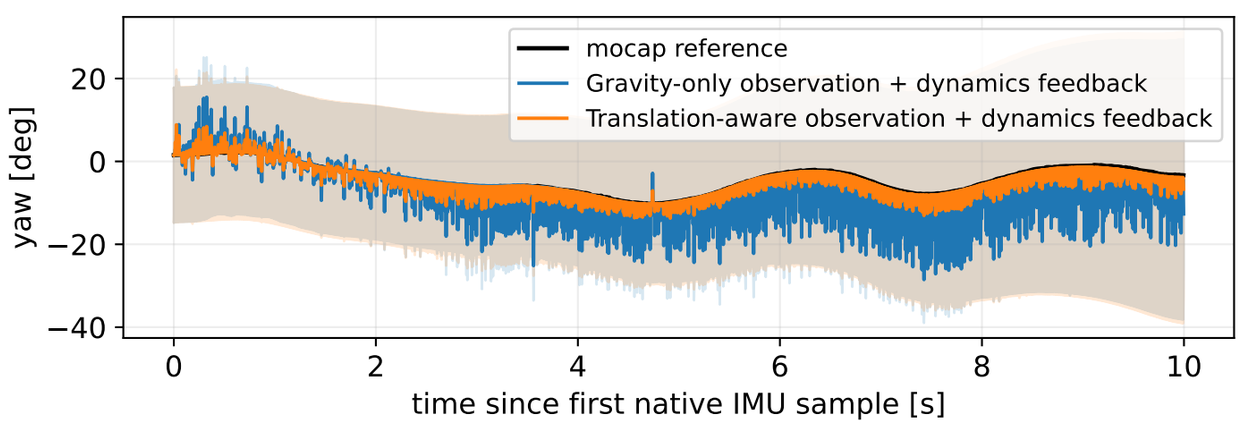




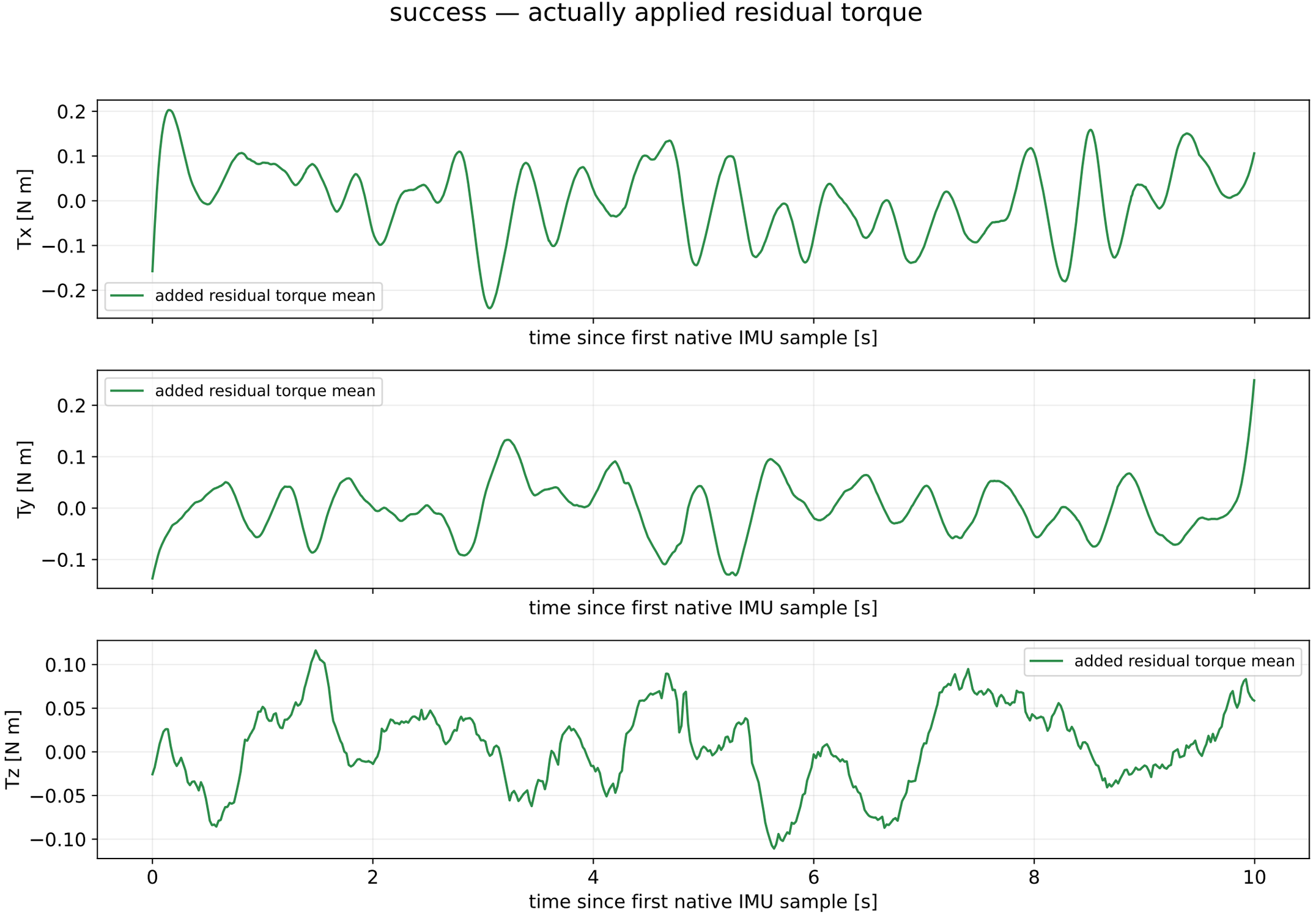








Attitude: IMU posterior-law local Euler approximation.
±2σ is not a 95% guarantee. Right column retains complete bands; black reference uses original mocap stamps.


success — actually applied residual torque
Tx [N m]
Ty [N m]
Tz [N m]
added residual torque mean
time since first native IMU sample [s]
0.2
0.1
0.0
−0.1
−0.2
0.10
0.05
0.00
−0.05
−0.10
0
2
4
6
8
10
Body coordinates, about the center of gravity; the same torque mean added to reconstruction dynamics.